Mohammad Ibrahim Saleem

+1 713-853-7974 - Ibrahimsaleem244@gmail.com | Houston, TX - Ibrahimsaleem.com - Portfolio - LinkedIn:Ibrahimsaleem91 - Git:ibrahimsaleem

Education

University of Houston Houston, TX
Master of Science in Cybersecurity *Expected August 2026*

- GPA: 4.0/4.0; Relevant Coursework: Network Security, Secure Enterprise Computing, Cryptography, Data Analysis for Cybersecurity, Risk Management

Rajiv Gandhi Proudyogiki Vishwavidyalaya
Bachelor of Technology in Computer Science Engineering *July 2023*

- Relevant Coursework: Python, Data Structures & Algorithms, DBMS, Cloud Computing, Operating Systems, Networking Protocols (TCP/IP, DNS, HTTP, SSL/TLS), Cryptography, Machine Learning

Experience

GenAI and Data Science Intern June 2025 – Present
NOV Inc (National Oilwell Varco) *Houston, TX (On-site)*

- Led the design and deployment of GenAI-based automation pipelines, reducing multi-day manual workflows to under 5 minutes, drastically accelerating internal operations.
- Improved automation accuracy from 92% to 98% by incorporating adaptive prompting, context-aware logic, and feedback-driven retry mechanisms. And compiling a research paper on this new approach.
- Developed and productionized a Modular Command & Processing (MCP) server to orchestrate autonomous AI agents, introducing self-healing capabilities for handling dynamic and inconsistent inputs.
- Built an end-to-end OCR and LLM pipeline for PDF data extraction, leveraging RAG architectures and integrating with Streamlit for real-time validation and interaction.
- Integrated AI workflows into Databricks and AWS, automating data ingestion, inference, logging, and evaluation.
- Enhanced internal AI tooling by deploying Streamlit-based GenAI dashboards, enabling non-technical users to leverage automation insights with minimal effort.
- Logged version-controlled results for continuous evaluation of LLM, OCR, and pipeline performance across a variety of document formats and business cases.

Research Assistant – AI & Cybersecurity — Git Repo Sep 2024 – May 2025
University of Houston *Houston, TX (On-site)*

- Led the design and implementation of LIMA as first author — a modular, LLM-driven penetration testing framework combining GPT-4o/Claude with an autonomous MCP-based system to perform initial machine access with minimal human input.
- Developed the PentestThinkingMCP server, integrating advanced reasoning agents with tools like Nmap, Metasploit, enum4linux, and BrowserMCP. Enabled real-time, context-aware vulnerability discovery and exploitation planning.
- Designed and executed the first quantitative benchmark comparing AI and human performance on HackTheBox machines; Claude 3.5 outperformed expert pentesters on multiple boxes with less then 0.05 run cost.
- Contributed novel insights into LLM reasoning weaknesses (e.g., GUI/tooling limitations) and adaptive failure points in real-world cybersecurity environments.
- Implemented a secure automotive IoT pairing mechanism using unencrypted TPMS signals and DSP techniques in Python and MATLAB, eliminating the need for manual password sharing.
- Synthesized findings from over 40 peer-reviewed papers to align research with current academic and industry trends.

Associate Software Engineer March 2023– February 2024
Nagarro Software Pvt. Ltd.

- Designed and developed scalable backend solutions using C#, .NET Core, and SQL Server, optimizing database queries to improve response times by 30%, and built secure REST APIs with JWT authentication and role-based access control to enhance security.
- Developed interactive Angular-based UI dashboards, improving operational efficiency and real-time data visualization, and integrated secure backend services to ensure seamless data communication.
- Automated data analytics pipelines using Python and Pandas, reducing manual reporting efforts by 40%, and implemented data-driven insights to improve business intelligence processes.

Research & Cyber Security Intern July 2022– December 2022
State Cyber Cell MP Police

- Conducted security assessments on critical infrastructure systems, identifying vulnerabilities in power grids and telecom networks, recommending risk mitigation strategies.
- Researched emerging attack vectors and applied Python-based threat intelligence analytics, improving incident response readiness
- Delivered cybersecurity awareness training to 500+ students and professionals on network security, cybercrime prevention

Projects

CyberPath AI – Personalized Learning Roadmaps | *Flask, GenAI,Python* Live

- Deployed an AI-driven web app that generates custom, interactive roadmaps for *Cybersecurity, AI Engineering, and Data Science.* Users click any topic to receive an on-demand explanation crafted by Gemini 2.0, dynamically tailored to their background, experience, and prior topics coverd.

PentestThinkingMCP – Autonomous Pen-Testing Server | *MCP, Beam Search, MCTS, AI Agent* Live MCP

- Designed an MCP Server which help LLM to *reasons, plans, and executes* complete attack chains using Beam Search + Monte Carlo Tree Search. Fully compromised HTB *"Lame"* in **3 min** end-to-end for ˜$0.03 in LLM cost.

LocalRAGAgent – Offline Retrieval-Augmented QA | *Python, LangChain, FastAPI, Chroma, Ollama* Repo

- Built a 100% local RAG pipeline that extracts text OCR from PDFs, embeds chunks into ChromaDB, retrieves context, and answers queries with Llama-3 via Ollama, delivering <300 ms latency on consumer hardware.

Secure Offline AI Assistant for Network Operators | *Huggingface, Flask, LLM* DataSet

- Fine-tuned a Llama 3.2 3b model via Unsloth on domain-specific logs, delivering on-prem NLP troubleshooting assistance and reducing mean-time-to-resolution by 30%.

Advanced Network Performance Monitoring System | *Python, ML, SDN (Ryu)* Repo

- Integrated ML-based DDoS detection (98% accuracy) with SDN controllers; automated real-time mitigation using MITRE ATT&CK tactics, improving network resilience.

Splunk-Based Detection & Response Automation | *Python, Splunk* Repo

- Built security dashboards and AbuseIPDB-powered reputation scoring that cut manual triage workload by 45%.

Cybersecurity Awareness Platform | *Python, JavaScript, Flask* Live

- Launched an interactive training suite featuring 30+ Python-built security labs; increased learner completion rates by 40%.

Skills & Certifications

Languages & Scripting: Python, JavaScript/TypeScript, C#, SQL, Bash/Shell, PowerShell

Frameworks & Libraries: Node.js, React, Angular, Flask, FastAPI, .NET Core, Streamlit, LangChain, PyTorch, TensorFlow, scikit-learn, Hugging Face, Llama/Ollama

Cloud, DevOps & MLOps: AWS (S3, Lambda, SageMaker, IAM, VPC), Azure, GCP, Docker, Kubernetes, Terraform, Jenkins, GitHub Actions, Databricks, CI/CD, static code analysis (Bandit, SonarQube)

Data & ML Engineering: Pandas, NumPy, ChromaDB, ETL/ELT pipelines, data warehousing, real-time analytics

Generative AI: LLM fine-tuning distillation (GPT-4o, Claude, Llama-3), prompt engineering, retrieval-augmented generation (RAG), autonomous agent orchestration (MCP, LangGraph), RLHF, synthetic data generation, model serving evaluation, Explainable AI

Databases: SQL Server, MySQL, PostgreSQL — design, tuning, optimization

Cybersecurity & Networking: Secure coding, Nmap, Metasploit, enum4linux, SDN (Ryu), Splunk, AbuseIPDB, MCP security agents, risk assessment incident response

Software Engineering: System design, distributed systems, RESTful APIs, TDD, debugging, automation pipelines, real-time data visualization

Professional: Research, Agile/Scrum collaboration, project management, technical communication

Certifications: AWS Certified Security – Specialty, CompTIA Security+, Microsoft Azure Security Fundamentals, Python for Networking Security, plus additional AI/ML MLOps credentials

Research

- **Saleem, M. I.** (First Author), Prabhakar, S. S., Harsha, A., Nagabhushan, D., Conklin, W. A., Lee, K. I., Banerjee, T. "*LIMA: Leveraging Large Language Models and MCP Servers for Initial Machine Access.*" *Proc. IEEE Intl. Conf. on Future Machine Learning and Data Science (FMLDS)*, 2025 (Complete Paper Submitted)

- NOV GenAI Team, **Saleem, M. I.** (Second Author). "*Self-Improving GenAI Agent for Fully Automated Report Parsing in Enterprise Environments.*" Research proposal submitted to the **Society of Petroleum Engineers (SPE)**, 2025 — work conducted as *GenAI & Data Science Intern, NOV Inc.*

- PHM Society Data Challenge 2025 – Working with NOV GenAI team to develop ML models for predicting aircraft engine maintenance events. Submission planned for PHM North America 2025 Conference.